

PKU GIMMS NDVI Description

README FILE AND SCRIPTS

by

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Data Set Version 1.2

Period: January 1982 to December 2022

Temporal Resolution: half month

Spatial Resolution: 1/12 degree

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1. DATA DESCRIPTION

The characteristic of the PKU GIMMS NDVI can be found in Table 1.

Table 1 Data description of PKU GIMMS NDVI.

Data Name	PKU GIMMS NDVI
Area	Global Vegetation area
Projection	Geographic
Spatial Resolution	1/12 degree
Temporal Resolution	Half month
Temporal Coverage	January 1982 to December 2022
Image Dimension	Rows: 2160 Columns: 4320
Units	Unitless
Fill Value	65535: Non-veg or $NDVI \leq 0$
Data Type	uint16
Valid Range	0-1000
Multiply By Scale Factor	0.001
File Format	TIFF(.tif)
File Size	~8Mb each file

2. FILE NAMING CONVENTION

PKU_GIMMS_NDVI_[version]_[year][month][half-month].[format]

Table 2 Description of file naming convention.

Variable	Description
PKU_GIMMS_NDVI	Abbreviation of Peking University GIMMS NDVI ● Constant
Version	Version_number ● Could be updated
Year	Digit year of the data ● From 1982 to 2022
Month	Digit month of the data ● From 01 to 12
Half month	Half month for the bimonthly dataset ● From 01 to 02, 01: the first half month; 02: the second half month
Format	Format of the data ● *.tif

For example, PKU_GIMMS_NDVI_V1.2_20010101 is the file name of PKU GIMMS NDVI image in the first half month of January in 2001.

3. QC LAYER DESCRIPTION.

Table 3 Description of QC layer.

Product Name	PKU GIMMS NDVI (AVHRR)	PKU GIMMS NDVI
Temporal coverage of data	1982-2015	1982-2022
Temporal coverage of QC layer	1982-2015	1982-2022
Data source	Solely based on AVHRR (1982-2015)	AVHRR (1982-2002) & The MODIS NDVI (2003-2022)
QC description	Inherited from the GIMMS NDVI3g product. 0: Good quality; 1: Spline interpolation; 2: Possible snow/cloud cover.	Constituted by three digits. The first digit represents the consolidation method, with: 1: Random forests ¹ ; 2: Linear function ² ; 3: Linear function ³ ; 4: Linear function ⁴ ; 5: Difference matching ⁵ . 9: N/A: 2003-2022. The second digit represents the QC of GIMMS NDVI before consolidation, with: 0: Good quality; 1: Spline interpolation; 2: Possible snow/cloud cover. 9: N/A: 2003-2022. The third digit represents the QC of MODIS NDVI, with: 0: Good Data; 1: Marginal data; 2: Snow/Ice; 3: Cloudy; 4: Estimated from MODIS historic time series; 9: N/A: 1982-2002.

Note:

¹Used when valid sample size>180 within the 11×11 window and the trained model had $R^2 \geq 0.1$, $RMSE \leq 0.1$, and $MAE \leq 0.1$. Three explanatory variables were involved, namely PKU GIMMS NDVI, latitude, longitude.

²Used when $30 \leq \text{valid sample size} < 180$ within the 11×11 window and the trained model had $R^2 \geq 0.1$, $\text{RMSE} \leq 0.1$, and $\text{MAE} \leq 0.1$. Three explanatory variables were involved, namely PKU GIMMS NDVI, normalized latitude, normalized longitude.

³Used when $2 \leq \text{valid sample size} < 30$ within the 11×11 window and the trained model had $R^2 \geq 0.1$, $\text{RMSE} \leq 0.1$, and $\text{MAE} \leq 0.1$. Single explanatory variable was used (PKU GIMMS NDVI).

⁴Used when $\text{valid sample size} < 2$ within the 11×11 window and the trained model had $R^2 \geq 0.1$, $\text{RMSE} \leq 0.1$, and $\text{MAE} \leq 0.1$. Single explanatory variable was used (PKU GIMMS NDVI).

⁵Used in all remaining situation.

4. HOW TO USE THE DATASETS IN MATLAB

Please use the `[importdata]` function to read the dataset. Here is an example:

```
clc
clear
% set the input dir
inpath = 'D:\file_folder\';
% set the temporal coverage
start_year = 1982;
end_year = 2022;
start_month = 1;
end_month = 12;

for year = start_year:1:end_year
    for month = start_month:1:end_month
        for half_month = 1:1:2
            % check the prefix
            prefix = 'PKU_GIMMS_NDVI_V1.2_';
            inname = strcat(inpath,prefix,num2str(year*10000+month*100+half_month),'.tif');
            % read data
            data = importdata(inname);
            data_ndvi = data(:, :, 1);
            data_QC = data(:, :, 2);
            data_QC = double(data_QC);
            data_QC (data_QC == 65535) = nan;
            data_ndvi = double(data_ndvi);
            data_ndvi(data_ndvi == 65535) = nan;
            data_ndvi = data_ndvi * 0.001;
            % Visually check the PKU GIMMS NDVI and QC
            figure(1); imagesc(data_ndvi);colorbar;
            figure(2); imagesc(data_QC);colorbar;
        end
    end
end
```